BASE CHANGE FOR THE RIGHT ADJOINT OF PUSHFORWARD UNDER TOR INDEPENDENCE, IN THREE CASES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0AA8 (sources/stacks-duality/pinned-original-source.tex line 1636).

Allowed standard prerequisites (editorial).

- The right adjoint a of pushforward, the base change diagram and its base change map. Same chapter, diagram labelled equation-base-change and equation labelled equation-base-change-map, invoked in the statement, and Lemma labelled lemma-iso-global-hom, cited at proof lines 1693-1697 and 1699-1703. Fixes the two functors and identifies both global Hom complexes with Hom on the base.
- Perfect generators of the derived category of quasi-coherent sheaves. Derived Categories of Schemes, Theorem labelled perfect-theorem-bondal-van-den-Bergh and Remark labelled perfect-remark-pullback-generator, cited at proof lines 1661-1666. Reduces the isomorphism of objects to the isomorphism of Hom complexes out of a generator.
- Base change for Hom out of a perfect object and comparison of the push-forwards. Derived Categories of Schemes, Lemmas labelled perfect-lemma-affine-morphism-and-hom-out-of-perfect and perfect-lemma-compare-base-change, cited at proof lines 1687-1691 and 1703-1707. Computes the left hand side and rewrites the right hand side.
- Base change for Hom over a ring, in the four cases. More on Algebra, Equation labelled more-algebra-equation-base-change-RHom, Section labelled more-algebra-section-base-change-RHom and Lemma labelled more-algebra-lemma-base-change-RHom parts (1) and (4), cited at proof lines 1713-1731 and 1819-1823. Says when beta is an isomorphism and identifies it with the map whose commutativity is proved at the end.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- The invertible transformation between the two compositions of functors. Same chapter, Lemma labelled lemma-flat-precompose-pus, cited at proof lines 1769-1773, and Derived Categories of Schemes, Remark labelled perfect-remark-multiplication-map, cited at proof lines 1787-1791. Supplies the maps mu and mu' whose commutativity reduces the final check to a triangle.

Source reference. Original equation 0A9J: sources/stacks-duality/pinned-original-source.tex, source lines 927–927. Same-chapter pointer into the author environment equation at source line 927; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original equation 0AA6: sources/stacks-duality/pinned-original-source.tex, source lines 1192–1192. Same-chapter pointer into the author environment equation at source line 1192; invoked at proof line 1666 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original theorem 09IS: sources/stacks-chow/tags.txt, source lines 12050–12050. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 1663 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0BQT: sources/stacks-chow/tags.txt, source lines 14746–14746. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 1665 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 0B6A: sources/stacks-chow/tags.txt, source lines 14063–14063. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 1675 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AA7: sources/stacks-chow/tags.txt, source lines 12975–12975. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 1690 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B6I: sources/stacks-duality/pinned-original-source.tex, source lines 878–878. Same-chapter pointer into the author environment lemma at source line 878; invoked at proof line 1693 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 08IB: sources/stacks-chow/tags.txt, source lines 10809–10809. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 1704 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 08D5: sources/stacks-chow/tags.txt, source lines 10625–10625. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 1708 of the counted proof. It is

retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original equation 0E1X: sources/stacks-chow/tags.txt, source lines 17587–17587. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 1717 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 06Z0: sources/stacks-chow/tags.txt, source lines 8904–8904. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 1719 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A6H: sources/stacks-chow/tags.txt, source lines 12845–12845. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 1720 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B91: sources/stacks-chow/tags.txt, source lines 14159–14159. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 1725 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A6A: sources/stacks-chow/tags.txt, source lines 12838–12838. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 1727 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B6G: sources/stacks-chow/tags.txt, source lines 14069–14069. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 1730 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 08E2: sources/stacks-chow/tags.txt, source lines 10658–10658. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 1735 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 08E8: sources/stacks-chow/tags.txt, source lines 10664–10664. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 1736 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9K: sources/stacks-duality/pinned-original-source.tex, source lines 957–957. Same-chapter pointer into the author environment lemma at source line 957; invoked at proof line 1774 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0BZA: sources/stacks-chow/tags.txt, source lines 15043–15043. Cross-chapter standard prerequisite located in the project tag

list staged in the local snapshot; invoked at proof line 1789 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 0E1V: sources/stacks-chow/tags.txt, source lines 17585–17585. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 1821 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In diagram (0A9J) assume*

(1) $g : Y' \to Y$ *is a morphism of affine schemes,*
(2) $f : X \to Y$ *is proper, and*
(3) f *and* g *are Tor independent.*

Then the base change map (0AA6) induces an isomorphism

$$L(g')^*a(K) \longrightarrow a'(Lg^*K)$$

in the following cases

(1) *for all* $K \in D_{QCoh}(\mathcal{O}_Y)$ *if* f *is flat of finite presentation,*
(2) *for all* $K \in D_{QCoh}(\mathcal{O}_Y)$ *if* f *is perfect and* Y *Noetherian,*
(3) *for* $K \in D^+_{QCoh}(\mathcal{O}_Y)$ *if* g *has finite Tor dimension and* Y *Noetherian.*

Proof. Write $Y = \mathrm{Spec}(A)$ and $Y' = \mathrm{Spec}(A')$. As a base change of an affine morphism, the morphism g' is affine. Let M be a perfect generator for $D_{QCoh}(\mathcal{O}_X)$, see Derived Categories of Schemes, Theorem 09IS. Then $L(g')^*M$ is a generator for $D_{QCoh}(\mathcal{O}_{X'})$, see Derived Categories of Schemes, Remark 0BQT. Hence it suffices to show that (0AA6) induces an isomorphism

$$R\mathrm{Hom}_{X'}(L(g')^*M, L(g')^*a(K)) \longrightarrow R\mathrm{Hom}_{X'}(L(g')^*M, a'(Lg^*K)) \tag{0.1.1}$$

of global hom complexes, see Cohomology, Section 0B6A, as this will imply the cone of $L(g')^*a(K) \to a'(Lg^*K)$ is zero. The structure of the proof is as follows: we will first show that these Hom complexes are isomorphic and in the last part of the proof we will show that the isomorphism is induced by (0.1.1).

The left hand side. Because M is perfect, the canonical map

$$R\mathrm{Hom}_X(M, a(K)) \otimes^{\mathbf{L}}_A A' \longrightarrow R\mathrm{Hom}_{X'}(L(g')^*M, L(g')^*a(K))$$

is an isomorphism by Derived Categories of Schemes, Lemma 0AA7. We can combine this with the isomorphism $R\mathrm{Hom}_Y(Rf_*M, K) = R\mathrm{Hom}_X(M, a(K))$ of Lemma 0B6I to get that the left hand side equals $R\mathrm{Hom}_Y(Rf_*M, K) \otimes^{\mathbf{L}}_A A'$.

The right hand side. Here we first use the isomorphism

$$R\mathrm{Hom}_{X'}(L(g')^*M, a'(Lg^*K)) = R\mathrm{Hom}_{Y'}(Rf'_*L(g')^*M, Lg^*K)$$

of Lemma 0B6I. Then we use the base change map $Lg^*Rf_*M \to Rf'_*L(g')^*M$ is an isomorphism by Derived Categories of Schemes, Lemma 08IB. Hence we may rewrite this as $R\mathrm{Hom}_{Y'}(Lg^*Rf_*M, Lg^*K)$. Since Y, Y' are affine and K, Rf_*M are in $D_{QCoh}(\mathcal{O}_Y)$ (Derived Categories of Schemes, Lemma 08D5) we have a canonical map

$$\beta : R\mathrm{Hom}_Y(Rf_*M, K) \otimes^{\mathbf{L}}_A A' \longrightarrow R\mathrm{Hom}_{Y'}(Lg^*Rf_*M, Lg^*K)$$

in $D(A')$. This is the arrow More on Algebra, Equation (0E1X) where we have used Derived Categories of Schemes, Lemmas 06Z0 and 0A6H to translate back and forth into algebra.

(1) If f is flat and of finite presentation, the complex Rf_*M is perfect on Y by Derived Categories of Schemes, Lemma 0B91 and β is an isomorphism by More on Algebra, Lemma 0A6A part (1).
(2) If f is perfect and Y Noetherian, the complex Rf_*M is perfect on Y by More on Morphisms, Lemma 0B6G and β is an isomorphism as before.
(3) If g has finite tor dimension and Y is Noetherian, the complex Rf_*M is pseudo-coherent on Y (Derived Categories of Schemes, Lemmas 08E2 and 08E8) and β is an isomorphism by More on Algebra, Lemma 0A6A part (4).

We conclude that we obtain the same answer as in the previous paragraph.

In the rest of the proof we show that the identifications of the left and right hand side of (0.1.1) given in the second and third paragraph are in fact given by (0.1.1). To make our formulas manageable we will use $(-,-)_X = R\mathrm{Hom}_X(-,-)$, use $-\otimes A'$ in stead of $-\otimes^{\mathbf{L}}_A A'$, and we will abbreviate $g^* = Lg^*$ and $f_* = Rf_*$. Consider the following commutative diagram

$$\begin{array}{ccccc}
((g')^*M, (g')^*a(K))_{X'} & \xleftarrow{\alpha} & (M, a(K))_X \otimes A' & = & (f_*M, K)_Y \otimes A' \\
\downarrow & & \downarrow & & \downarrow \\
((g')^*M, (g')^*a(g_*g^*K))_{X'} & \xleftarrow{\alpha} & (M, a(g_*g^*K))_X \otimes A' & = & (f_*M, g_*g^*K)_Y \otimes A' \\
\uparrow & & \uparrow & \overset{\mu'}{\swarrow} & \\
((g')^*M, (g')^*g'_*a'(g^*K))_{X'} & \xleftarrow{\alpha} & (M, g'_*a'(g^*K))_X \otimes A' & & (f_*M, K) \otimes A' \\
\downarrow & \overset{\mu}{\swarrow} & & & \downarrow \beta \\
((g')^*M, a'(g^*K))_{X'} & = & (f'_*(g')^*M, g^*K)_{Y'} & \longrightarrow & (g^*f_*M, g^*K)_{Y'}
\end{array}$$

The arrows labeled α are the maps from Derived Categories of Schemes, Lemma 0AA7 for the diagram with corners X', X, Y', Y. The upper part of the diagram is commutative as the horizontal arrows are functorial in the entries. The middle vertical arrows come from the invertible transformation $g'_* \circ a' \to a \circ g_*$ of Lemma 0A9K and therefore the middle square is commutative. Going down the left hand side is (0.1.1). The upper horizontal arrows provide the identifications used in the second paragraph of the proof. The lower horizontal arrows including β provide the identifications used in the third paragraph of the proof. Given $E \in D(A)$, $E' \in D(A')$, and $c : E \to E'$ in $D(A)$ we will denote $\mu_c : E \otimes A' \to E'$ the map induced by c and the adjointness of restriction and base change; if c is clear we write $\mu = \mu_c$, i.e., we drop c from the notation. The map μ in the diagram is of this form with c given by the identification $(M, g'_*a(g^*K))_X = ((g')^*M, a'(g^*K))_{X'}$; the triangle involving μ is commutative by Derived Categories of Schemes, Remark 0BZA.

Observe that

$$\begin{array}{ccccc}
(M, a(g_*g^*K))_X & = & (f_*M, g_*g^*K)_Y & = & (g^*f_*M, g^*K)_{Y'} \\
\uparrow & & & & \uparrow \\
(M, g'_*a'(g^*K))_X & = & ((g')^*M, a'(g^*K))_{X'} & = & (f'_*(g')^*M, g^*K)_{Y'}
\end{array}$$

is commutative by the very definition of the transformation $g'_* \circ a' \to a \circ g_*$. Letting μ' be as above corresponding to the identification $(f_*M, g_*g^*K)_X = (g^*f_*M, g^*K)_{Y'}$, then the hexagon commutes as well. Thus it suffices to show that β is equal to the

composition of $(f_*M, K)_Y \otimes A' \to (f_*M, g_*g^*K)_X \otimes A'$ and μ'. To do this, it suffices to prove the two induced maps $(f_*M, K)_Y \to (g^*f_*M, g^*K)_{Y'}$ are the same. In other words, it suffices to show the diagram

$$\begin{array}{ccc} R\mathrm{Hom}_A(E, K) & \xrightarrow[\text{induced by } \beta]{} & R\mathrm{Hom}_{A'}(E \otimes^{\mathbf{L}}_A A', K \otimes^{\mathbf{L}}_A A') \\ & \searrow \quad \nearrow & \\ & R\mathrm{Hom}_A(E, K \otimes^{\mathbf{L}}_A A') & \end{array}$$

commutes for all $E, K \in D(A)$. Since this is how β is constructed in More on Algebra, Section 0E1V the proof is complete. $\square$